

SHALINI BANGAMUWAGE

IT Undergraduate — Full-Stack Developer

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PROFILE

High-achieving Information Technology undergraduate at the University of Moratuwa (3.84 GPA), focused on Software Engineering and full-stack development. Hands-on experience building and deploying web, mobile, and software applications using React, Node.js, NestJS, Next.js, and modern databases. Strong foundation in OOP, data structures & algorithms, REST APIs, database design, authentication, and scalable software development. Built 10 projects spanning full-stack web, mobile, AI-powered, and embedded systems, integrating external APIs into production-ready solutions.

EDUCATION

University of Moratuwa — Faculty of Information Technology — GPA: 3.84

Undergraduate, BSc (Hons) in Information Technology (2024–2028)

St. Thomas' Girls' High School (2014–2022)

- G.C.E. A/L: Physical Science stream — A, A, B (2022); District Rank 142; Z-Score 1.82

SKILLS

- Languages: Java, C/C++, JavaScript, TypeScript, Python, Go
- Frontend: React, React Native (Expo), Next.js, Tailwind CSS
- Backend: Node.js, Express, NestJS, FastAPI
- Databases: MongoDB, Firebase/Firestore, Supabase (Postgres), PostgreSQL, pgvector, Redis
- Cloud, DevOps & Testing: AWS, Docker, GitHub Actions, Git, Playwright, axe-core, Agile, Scrum
- APIs & Integrations: REST API design, Groq API, Google OAuth, PayHere, Cloudinary, Meilisearch
- Embedded & Hardware: Arduino, ESP32, VS1053B, embedded systems
- Software Engineering Fundamentals: Object-Oriented Programming (OOP), Data Structures & Algorithms, SQL, Database Design, Authentication, Software Testing

PROJECTS

MediCareX — Pharmacy Management System — React (Vite), NestJS, Firebase/Firestore, Cloudinary

GitHub: github.com/SoftwareProjectInnovateX | Live: innovatexuom.vercel.app/

- Built a responsive frontend with React, Vite, and Tailwind CSS, featuring reusable UI components for 2 user roles — customers and admins.
- Developed a NestJS and Firebase backend spanning 4 core modules — authentication, order management, prescription processing, and product catalog — with role-based access for 4 user types.

PaperPulse — AI-Powered Research Paper Analysis Tool — Next.js, Supabase, Groq API, Semantic Scholar API

GitHub: github.com/ShaliniBangamuwage/paperpulse | Live: paperpulse-taupe.vercel.app

- Created a research platform with 4 core features — paper upload, literature discovery, AI chat, and project idea generation with roadmaps.
- Integrated 4 external services — Groq AI, PDF parsing, Stripe, and Resend — for content generation, billing, and real-time data handling.

HabitAutopsy — AI Habit-Tracking App — React Native (Expo), Node.js, Express, Supabase, Groq API

GitHub: github.com/ShaliniBangamuwage/habit_tracker | Live: habittracker-shalini-madhukas-projects.vercel.app/

- Built a full-stack habit tracker across 4 layers — React Native/Expo frontend, TypeScript backend, Prisma ORM, and Supabase database.
- Delivered 3 AI-powered features — coaching, relapse support, and analytics — for personalized wellness tracking across mobile and backend systems.

MathsLanka — Government School Learning Platform — Next.js, React, TypeScript, Tailwind CSS, NestJS, Firebase, Turborepo

GitHub: github.com/ShaliniBangamuwage/Government_school | Live: edunexa-school-platform.vercel.app/

- Engineered **MathsLanka**, a full-stack educational platform using Next.js, NestJS, Firebase, and Turborepo, supporting workflows for 3 dashboards — teacher, student, and admin.
- Implemented 5 core features — curriculum access, quizzes, math simulators, AI tutoring, and brain games — plus role-based access control.

PipelineMedic — AI-Powered CI/CD Failure Triage Platform — React, TypeScript, FastAPI, Python, PostgreSQL, GitHub API, LLM APIs

GitHub: github.com/ShaliniBangamuwage/PipelineMedic

- Developed a platform with 3 core functions — analyzing CI/CD failure logs, identifying root causes, and recommending remediation steps.
- Implemented 5 platform features — GitHub integration, multi-tenant authentication, background jobs, PR comments, and secure patch suggestions.

GateFlow — Multi-Tenant API Gateway & Rate Limiter (In Development) — Go, React, TypeScript, PostgreSQL, Redis, Docker

GitHub: github.com/ShaliniBangamuwage/GateFlow

- Building a multi-tenant API gateway with 3 core features — API-key authentication, upstream request routing, and token-bucket rate limiting.
- Developing an administrative dashboard with 3 functions — configuring clients, monitoring API traffic, and analyzing rejected requests.

AccessiScan — Automated Accessibility Auditing Platform (In Development) — React, TypeScript, Node.js, PostgreSQL, Playwright, Axe-core

GitHub: github.com/ShaliniBangamuwage/AccessiScan

- Building an automated platform that scans websites for 3 WCAG issue types — missing alt text, poor contrast, and unlabeled forms.
- Implementing 4 reporting features — severity-based reports, remediation guidance, scan comparisons, and improvement tracking.

Restaurant E-Commerce Website — React, Tailwind CSS, Node.js, Express, MongoDB, Mongoose

GitHub: github.com/ShaliniBangamuwage/cafe-fullstack-app

- Delivered a full-stack restaurant ordering platform with a React/Tailwind frontend, Express backend, and a 3-collection MongoDB schema for menu items, customers, and orders.

Laser Tune — Embedded Musical Device — Arduino Mega, ESP32, C++, VS1053B, Embedded Systems

GitHub: github.com/kevizamarz/LaserTunesCode

- Engineered a musical device integrating 3 hardware components — Arduino Mega, ESP32, VS1053B audio codec — with custom C++ firmware processing 2 real-time sensor inputs (piezo, FSR).

VOLUNTEER EXPERIENCE

Director of Membership Development & Volunteer Coordination — IEEE WIE Student Branch Affinity Group, University of Moratuwa

- Lead membership engagement and outreach across 4 IEEE WIE focus areas.
- Co-Chair Enginuity 3.0 (2026), overseeing planning, logistics, volunteers, and execution.
- Manage 4 areas of IEEE IAS logistics — venues, schedules, equipment, volunteers.
- Supported MERCon, Open Week, Robotics Day, and Hackelite operations.

CERTIFICATION

- SQL Server for Data Analysis — Alison, Aug 2025 | ID: 5603-46531290
- Diploma in HTML & CSS Web Development — Alison, Feb 2025 | ID: 3972-46531290
- Python Zero-to-Hero — Udemy, Feb 2025 | ID: UC-11e166a52b3c4302b5646ad86e6b54e5
- Career Essentials in Software Development — Microsoft & LinkedIn, Feb 2025
- AWS Cloud Essentials Knowledge Badge Readiness Path with Labs — AWS Training & Certification

ACHIEVEMENTS

- Winner, UI/UX Prototype Phase — AlgoArena (Leo Club, University of Sri Jayewardenepura), 2025 — contributed Figma prototyping as part of Team NEONFRAME
- Selected to exhibit LaserTunes (embedded musical device) at FITExpo 2025, Faculty of Information Technology, University of Moratuwa